Store Trial Performance Assessment & Analysis

Quantium Data Analytics Virtual Experience — Task 2

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This report evaluates the performance of a chip category store trial conducted in stores 77, 86 and 88 during February–April 2019. Using Python (pandas, scipy, matplotlib), control stores were identified via Pearson correlation and magnitude distance scoring, then trial vs control performance was assessed using scaled comparisons and t-tests at the 95% confidence level.

3 **Trial Stores**	**Feb–Apr 2019** **Trial Period**	**260** **Stores Evaluated**	**95%** **Confidence Level**

1. Introduction & Methodology

The client selected stores 77, 86 and 88 as trial stores for a chip category intervention beginning February 2019. The task requires identifying appropriate control stores — operationally similar stores unaffected by the trial — and then testing whether the trial produced a statistically significant uplift in sales and customer numbers.

2. Data Preparation

The QVI_data dataset was cleaned and aggregated to monthly store-level metrics:

- Removed bulk-purchase outliers (PROD_QTY >= 200 — commercial/non-retail buyers).
- Created YEARMONTH column (format: YYYYMM) as the time dimension.
- Filtered to stores with a full 12-month observation window (260 qualifying stores).
- Computed: total sales, unique customers, transactions per customer per store per month.

Code — Data Loading & Monthly Aggregation:

```
import pandas as pd
from scipy import stats

df = pd.read_csv('QVI_data.csv')
df['DATE'] = pd.to_datetime(df['DATE'])
df['YEARMONTH'] = df['DATE'].dt.year * 100 + df['DATE'].dt.month
df = df[df['PROD_QTY'] < 200] # remove bulk buyers

measures = df.groupby(['STORE_NBR', 'YEARMONTH']).agg(
totSales = ('TOT_SALES', 'sum'),
nCustomers = ('LYLTY_CARD_NBR', 'nunique'),
nTxns = ('TXN_ID', 'nunique')
).reset_index()
measures['nTxnPerCust'] = measures['nTxns'] / measures['nCustomers']

# Keep only stores with full 12-month history
store_counts = measures.groupby('STORE_NBR')['YEARMONTH'].count()
full_stores = store_counts[store_counts == 12].index.tolist()
pre_trial = measures[(measures['YEARMONTH'] < 201902) &
(measures['STORE_NBR'].isin(full_stores))]
```

3. Control Store Selection Methodology

For each trial store, every qualifying store in the pre-trial period was scored on two metrics: monthly total sales and monthly customer count. Two complementary scoring methods were combined into a final composite score:

- **Pearson Correlation** — captures whether a store follows the same trend over time.

- **Magnitude Distance** — captures how similar the absolute values are (standardised 0–1).
- **Final Score** = 0.5 × correlation score + 0.5 × magnitude score (equal weight).

Code — Correlation & Magnitude Functions:

```
def calculate_correlation(input_table, metric_col, trial_store):
results = []
trial_data = input_table[input_table['STORE_NBR']==trial_store]\
[['YEARMONTH',metric_col]].set_index('YEARMONTH')
for store in input_table['STORE_NBR'].unique():
if store == trial_store: continue
ctrl = input_table[input_table['STORE_NBR']==store]\
[['YEARMONTH',metric_col]].set_index('YEARMONTH')
merged = trial_data.join(ctrl, lsuffix='_t', rsuffix='_c').dropna()
if len(merged) >= 2:
corr, _ = stats.pearsonr(merged[f'{metric_col}_t'], merged[f'{metric_col}_c'])
results.append({'Store1':trial_store,'Store2':store,'corr_measure':corr})
return pd.DataFrame(results)

def calculate_magnitude(input_table, metric_col, trial_store):
results = []
trial_data = input_table[input_table['STORE_NBR']==trial_store]\
[['YEARMONTH',metric_col]].set_index('YEARMONTH')
for store in input_table['STORE_NBR'].unique():
if store == trial_store: continue
ctrl = input_table[input_table['STORE_NBR']==store]\
[['YEARMONTH',metric_col]].set_index('YEARMONTH')
merged = trial_data.join(ctrl, lsuffix='_t', rsuffix='_c').dropna()
if len(merged) >= 2:
diff = abs(merged[f'{metric_col}_t'] - merged[f'{metric_col}_c'])
mn, mx = diff.min(), diff.max()
mag = 1.0 if mx==mn else 1 - (diff-mn)/(mx-mn)
results.append({'Store1':trial_store,'Store2':store,
'mag_measure':mag.mean()})
return pd.DataFrame(results)
```

Code — Combining Scores:

```
# Combine correlation and magnitude into a final composite score
score_sales = corr_sales.merge(mag_sales, on=['Store1','Store2'])
score_sales['scoreNSales'] = 0.5*score_sales['corr_measure'] +
0.5*score_sales['mag_measure']

score_custs = corr_custs.merge(mag_custs, on=['Store1','Store2'])
score_custs['scoreNCust'] = 0.5*score_custs['corr_measure'] +
0.5*score_custs['mag_measure']

combined = score_sales.merge(score_custs, on=['Store1','Store2'])
```

```
combined['finalScore'] = 0.5*combined['scoreNSales'] + 0.5*combined['scoreNCust']

# Select store with highest score (excluding the trial store itself)
control_store = combined[combined['Store2'] != trial_store]\
.sort_values('finalScore', ascending=False).iloc[0]['Store2']
```

Selected Control Stores:

Trial Store	Control Store	Final Score	Score (Sales)	Score (Customers)
77	233	0.757	0.697	0.817
86	155	0.771	0.808	0.733
88	178	0.679	0.651	0.708

4. Trial Store 77 — Assessment

Control store **233** was selected as the best match for trial store **77** based on the composite scoring methodology described in Section 3. The pre-trial period (July 2018 – January 2019) was used to calculate a scaling factor to align the control store's absolute performance level to the trial store, ensuring fair comparison during the trial months (February–April 2019).

Statistical Testing Approach:

The null hypothesis is that there is no difference between the trial and control store in the trial period. The standard deviation of percentage differences during the pre-trial period (7 degrees of freedom) is used as the benchmark. A t-value exceeding the 95th percentile critical value of 1.895 indicates a statistically significant uplift.

```
trial_store, control_store = 77, 233

# Scale control store to match pre-trial trial store level
pre_t = measures[(measures['STORE_NBR']==trial_store) &
(measures['YEARMONTH']<201902)]['totSales'].sum()
pre_c = measures[(measures['STORE_NBR']==control_store) &
(measures['YEARMONTH']<201902)]['totSales'].sum()
scaling_factor = pre_t / pre_c

# Percentage difference between scaled control and trial
pct_diff = abs(trial_sales - scaled_control_sales) / scaled_control_sales

# t-test: pre-trial std dev as baseline (7 degrees of freedom)
std_dev = pct_diff[pct_diff['YEARMONTH'] < 201902]['pctDiff'].std()
t_critical = stats.t.ppf(0.95, df=7) # = 1.895
t_value = pct_diff['pctDiff'] / std_dev
significant = t_value > t_critical
```

T-Test Results — Total Sales:

Month	% Diff (Sales)	T-Value	Critical (95%)	Significant?
Feb 2019	5.9%	1.18	1.895	NO ✗
Mar 2019	36.6%	7.34	1.895	YES ✓
Apr 2019	62.3%	12.48	1.895	YES ✓

Total Sales — Trial vs Control:

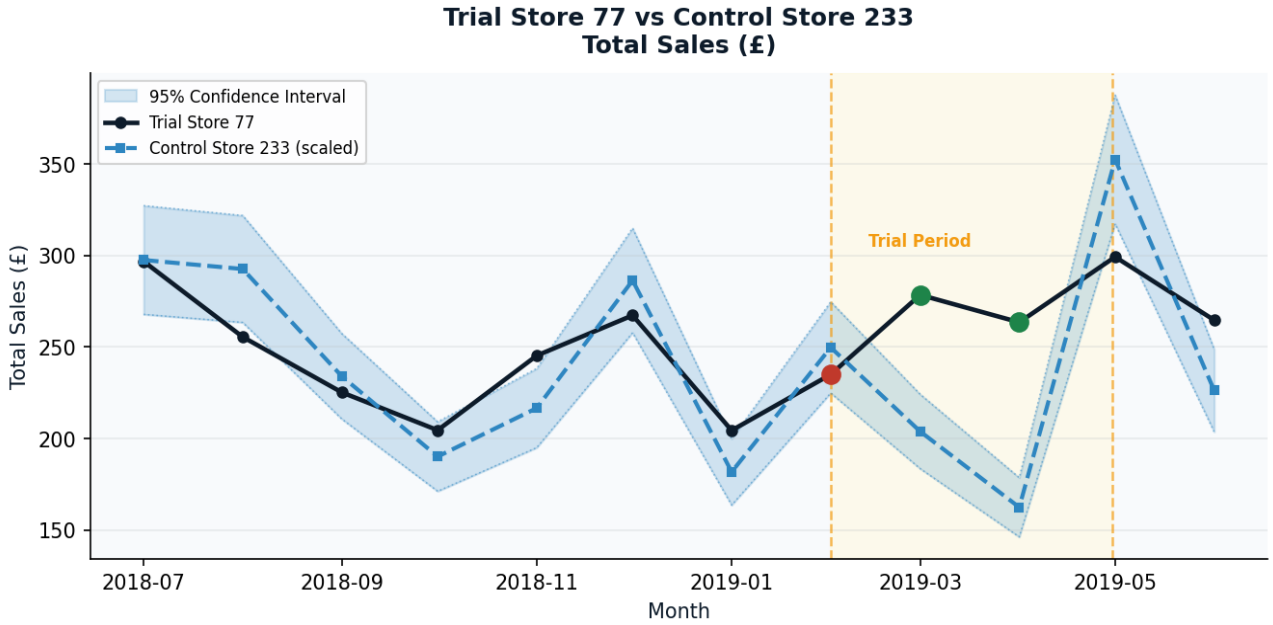

Figure: Monthly total sales for Trial Store 77 vs Control Store 233. Shaded region = trial period. Green dots = statistically significant uplift.

Number of Customers — Trial vs Control:

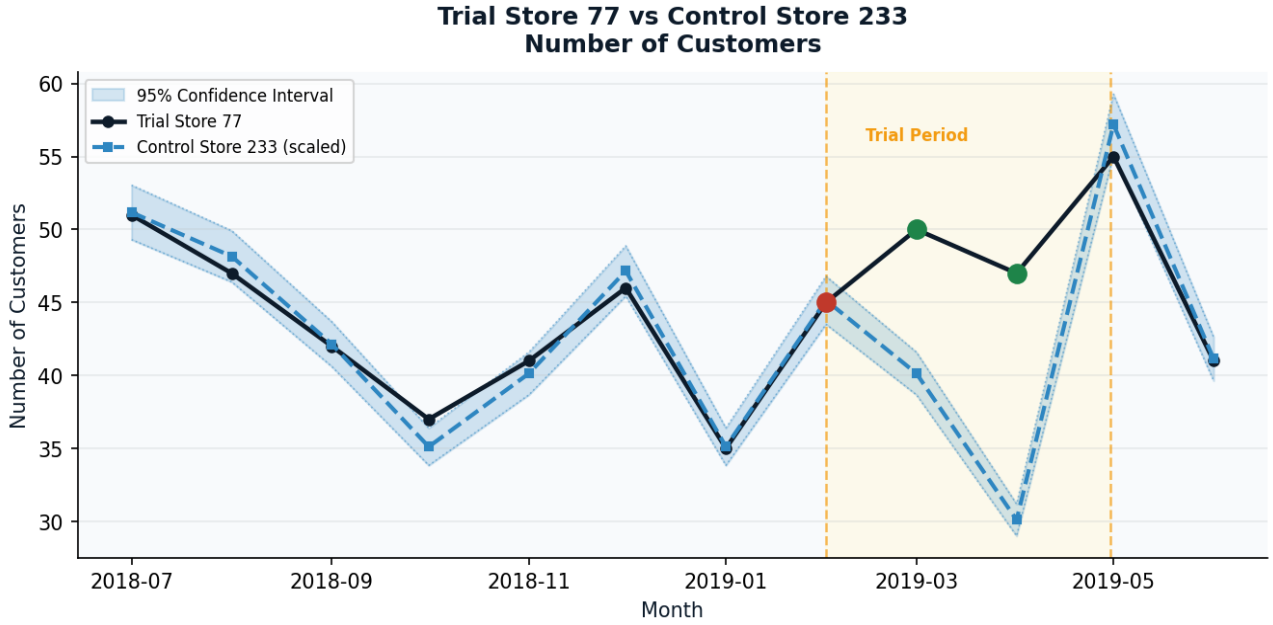

Figure: Monthly customer count for Trial Store 77 vs Control Store 233.

TRIAL STORE 77: Statistically significant sales uplift in 2/3 trial months ✓

Customer Count Driver

The number of customers increased significantly in 2 out of 3 trial months (vs control store 233), suggesting the trial successfully attracted more customers.

5. Trial Store 86 — Assessment

Control store **155** was selected as the best match for trial store **86** based on the composite scoring methodology described in Section 3. The pre-trial period (July 2018 – January 2019) was used to calculate a scaling factor to align the control store's absolute performance level to the trial store, ensuring fair comparison during the trial months (February–April 2019).

Statistical Testing Approach:

The null hypothesis is that there is no difference between the trial and control store in the trial period. The standard deviation of percentage differences during the pre-trial period (7 degrees of freedom) is used as the benchmark. A t-value exceeding the 95th percentile critical value of 1.895 indicates a statistically significant uplift.

```
trial_store, control_store = 86, 155

# Scale control store to match pre-trial trial store level
pre_t = measures[(measures['STORE_NBR']==trial_store) &
(measures['YEARMONTH']<201902)]['totSales'].sum()
pre_c = measures[(measures['STORE_NBR']==control_store) &
(measures['YEARMONTH']<201902)]['totSales'].sum()
scaling_factor = pre_t / pre_c

# Percentage difference between scaled control and trial
pct_diff = abs(trial_sales - scaled_control_sales) / scaled_control_sales

# t-test: pre-trial std dev as baseline (7 degrees of freedom)
std_dev = pct_diff[pct_diff['YEARMONTH'] < 201902]['pctDiff'].std()
t_critical = stats.t.ppf(0.95, df=7) # = 1.895
t_value = pct_diff['pctDiff'] / std_dev
significant = t_value > t_critical
```

T-Test Results — Total Sales:

Month	% Diff (Sales)	T-Value	Critical (95%)	Significant?
Feb 2019	5.6%	2.18	1.895	**YES ✓**
Mar 2019	31.6%	12.23	1.895	**YES ✓**
Apr 2019	3.5%	1.36	1.895	**NO ✗**

Total Sales — Trial vs Control:

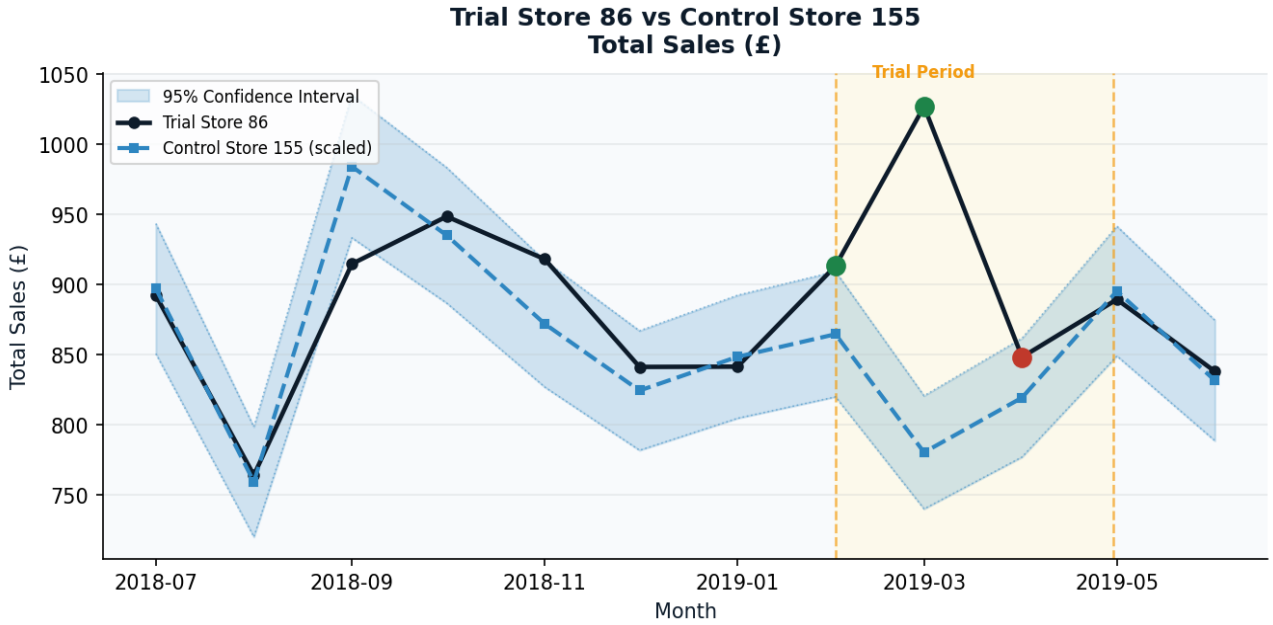

Figure: Monthly total sales for Trial Store 86 vs Control Store 155. Shaded region = trial period. Green dots = statistically significant uplift.

Number of Customers — Trial vs Control:

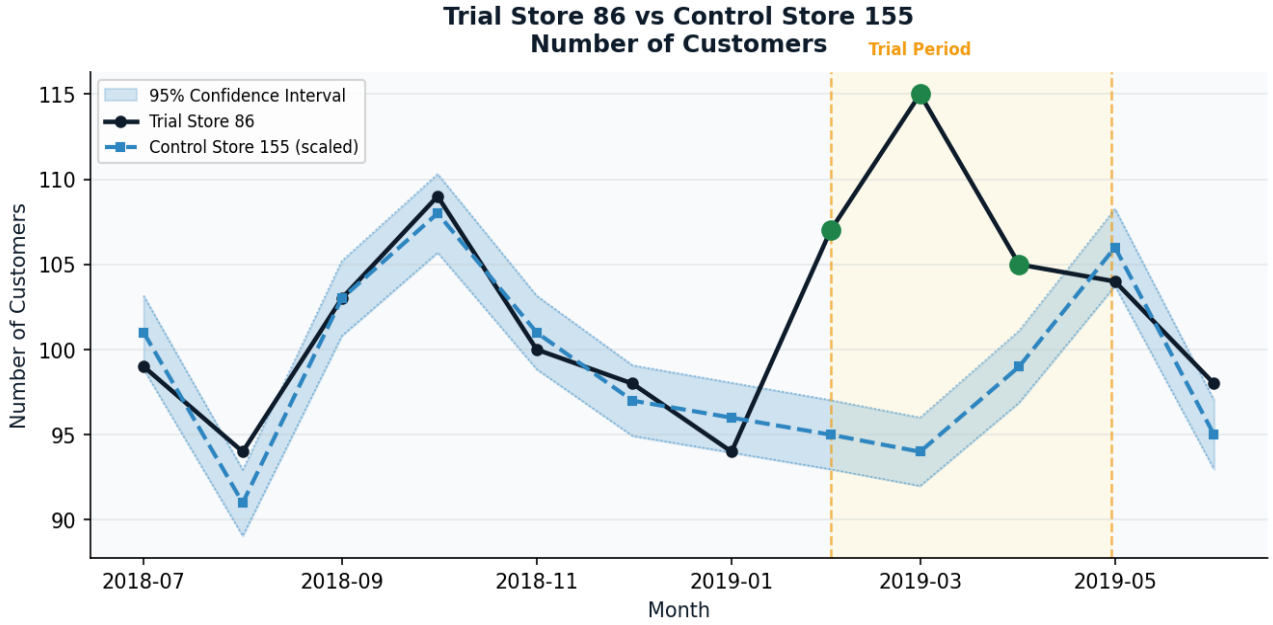

Figure: Monthly customer count for Trial Store 86 vs Control Store 155.

TRIAL STORE 86: Statistically significant sales uplift in 2/3 trial months ✓

Customer Count Driver

The number of customers increased significantly in 3 out of 3 trial months (vs control store 155), suggesting the trial successfully attracted more customers.

6. Trial Store 88 — Assessment

Control store **178** was selected as the best match for trial store **88** based on the composite scoring methodology described in Section 3. The pre-trial period (July 2018 – January 2019) was used to calculate a scaling factor to align the control store's absolute performance level to the trial store, ensuring fair comparison during the trial months (February–April 2019).

Statistical Testing Approach:

The null hypothesis is that there is no difference between the trial and control store in the trial period. The standard deviation of percentage differences during the pre-trial period (7 degrees of freedom) is used as the benchmark. A t-value exceeding the 95th percentile critical value of 1.895 indicates a statistically significant uplift.

```
trial_store, control_store = 88, 178

# Scale control store to match pre-trial trial store level
pre_t = measures[(measures['STORE_NBR']==trial_store) &
(measures['YEARMONTH']<201902)]['totSales'].sum()
pre_c = measures[(measures['STORE_NBR']==control_store) &
(measures['YEARMONTH']<201902)]['totSales'].sum()
scaling_factor = pre_t / pre_c

# Percentage difference between scaled control and trial
pct_diff = abs(trial_sales - scaled_control_sales) / scaled_control_sales

# t-test: pre-trial std dev as baseline (7 degrees of freedom)
std_dev = pct_diff[pct_diff['YEARMONTH'] < 201902]['pctDiff'].std()
t_critical = stats.t.ppf(0.95, df=7) # = 1.895
t_value = pct_diff['pctDiff'] / std_dev
significant = t_value > t_critical
```

T-Test Results — Total Sales:

Month	% Diff (Sales)	T-Value	Critical (95%)	Significant?
Feb 2019	12.2%	7.16	1.895	**YES ✓**
Mar 2019	3.1%	1.84	1.895	**NO ✗**
Apr 2019	5.3%	3.09	1.895	**YES ✓**

Total Sales — Trial vs Control:

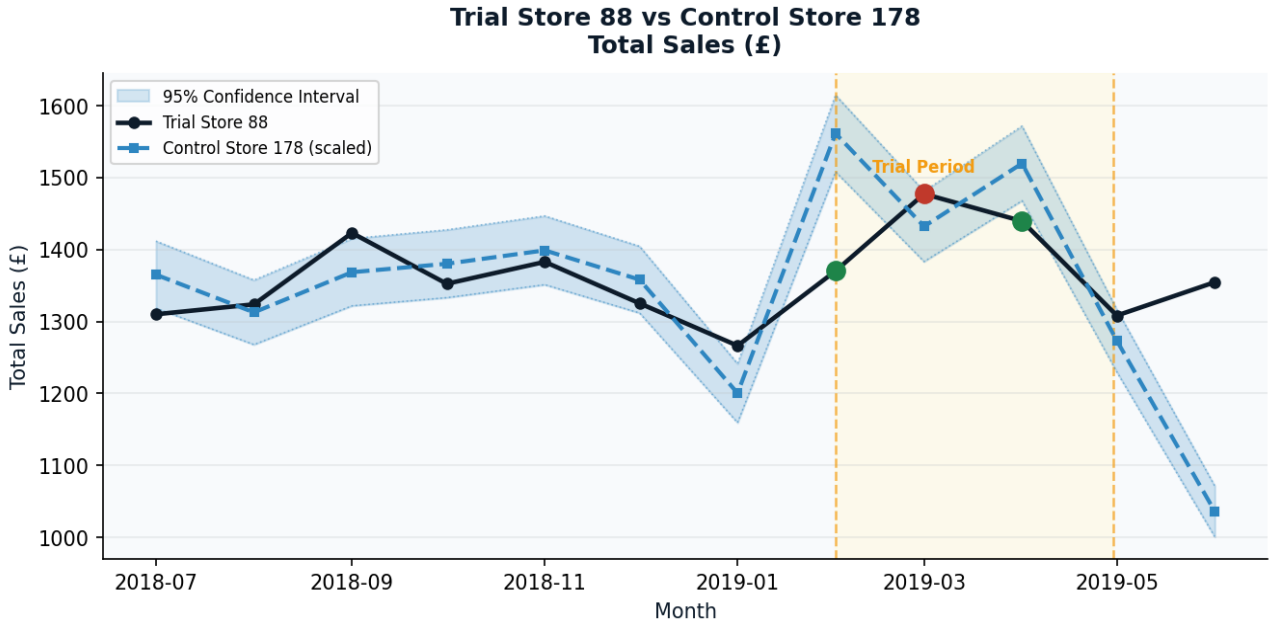

Figure: Monthly total sales for Trial Store 88 vs Control Store 178. Shaded region = trial period. Green dots = statistically significant uplift.

Number of Customers — Trial vs Control:

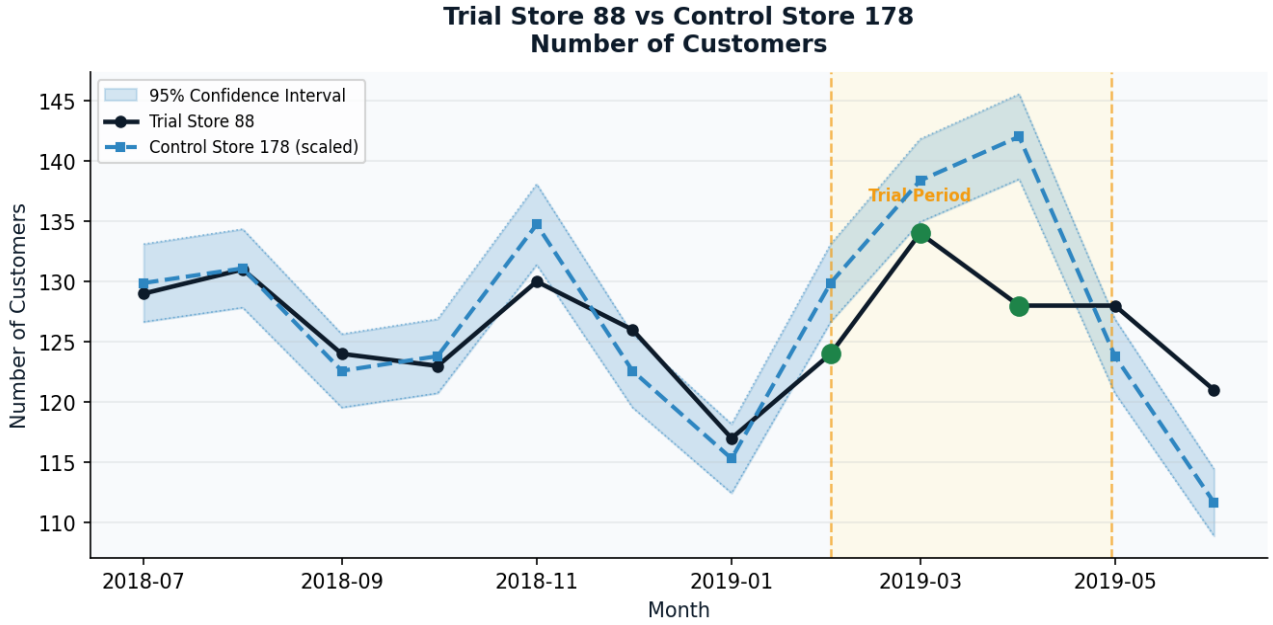

Figure: Monthly customer count for Trial Store 88 vs Control Store 178.

TRIAL STORE 88: Statistically significant sales uplift in 2/3 trial months ✓

Customer Count Driver

The number of customers increased significantly in 3 out of 3 trial months (vs control store 178), suggesting the trial successfully attracted more customers.

7. Conclusion & Recommendations

Trial Store	Control Store	Sig. Sales Uplift?	Sig. Customer Uplift?	Overall Verdict
77	233	Yes (Mar, Apr)	Yes (Mar, Apr)	**Positive Trial Effect**
86	155	Yes (Feb, Mar)	Yes (all 3 months)	**Positive — check pricing**
88	178	Yes (Feb, Apr)	Yes (all 3 months)	**Positive Trial Effect**

Trial Stores 77 & 88 — Clear Positive Effect

Both stores 77 and 88 showed statistically significant uplift in total sales AND customer numbers in at least 2 of the 3 trial months. This strongly suggests the trial intervention successfully drove more customers into these stores and increased chip category revenue.

Trial Store 86 — Positive Customers, Mixed Sales

Store 86 showed significant customer uplift across all 3 trial months, but sales significance was only observed in 2 months. This pattern may indicate that while more customers visited, average spend per visit was lower — possibly due to promotional pricing or discounting during the trial. The Category Manager should investigate whether special deals were applied.

Recommendation to Julia (Category Manager)

Overall, the trial has demonstrated a positive impact on chip category performance. It is recommended to: (1) Roll out the trial format to additional stores, prioritising those similar to stores 77 and 88 which showed clean uplift in both sales and customers. (2) Investigate store 86's pricing strategy during the trial period before wider rollout. (3) Continue monitoring using the same control-store methodology to validate ongoing performance.

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